

STM32-Based Multisine EIS Platform for Equivalent-Circuit Parameter Estimation

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Overview

Introduction

System Architecture Overview

Hardware and Software

IS Measurement Results

GUI & Dataset Pipeline

Machine Learning Model Design

Results and Evaluation

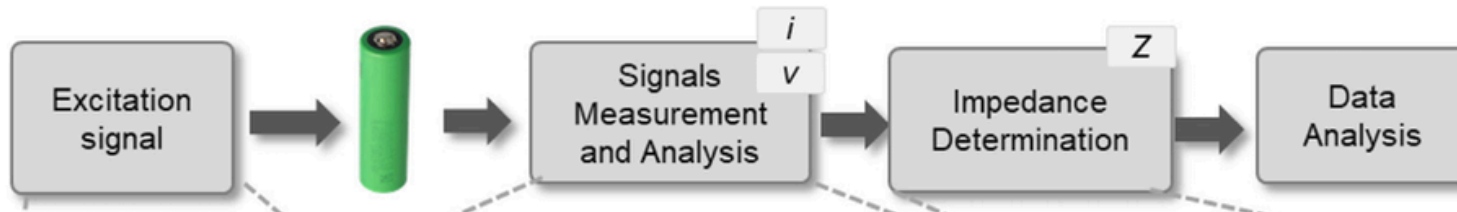
Limitations & Conclusion



Member Responsibilities

Member	Presentation
Gia Khanh Nguyen	Introduction System Architecture Overview Hardware and Software
Minh Quang Vu	IS Measurement Results
Khoi Nguyen Nguyen	GUI & Dataset Pipeline Machine Learning Model Design
Tuan Minh Le	Results and Evaluation Limitations & Conclusion

Introduction



"EIS: characterizing a material or device by measuring impedance across a frequency range"

THE PROBLEM



Expensive & bulky

Commercial impedance analyzers cost tens of thousands of euros and require the DUT to be brought into a lab.



Expert-driven fitting

Interpreting a spectrum means offline curve-fitting to an equivalent circuit, supervised by a specialist.

TWO QUESTIONS TO ANSWER



Q1 — Measurement

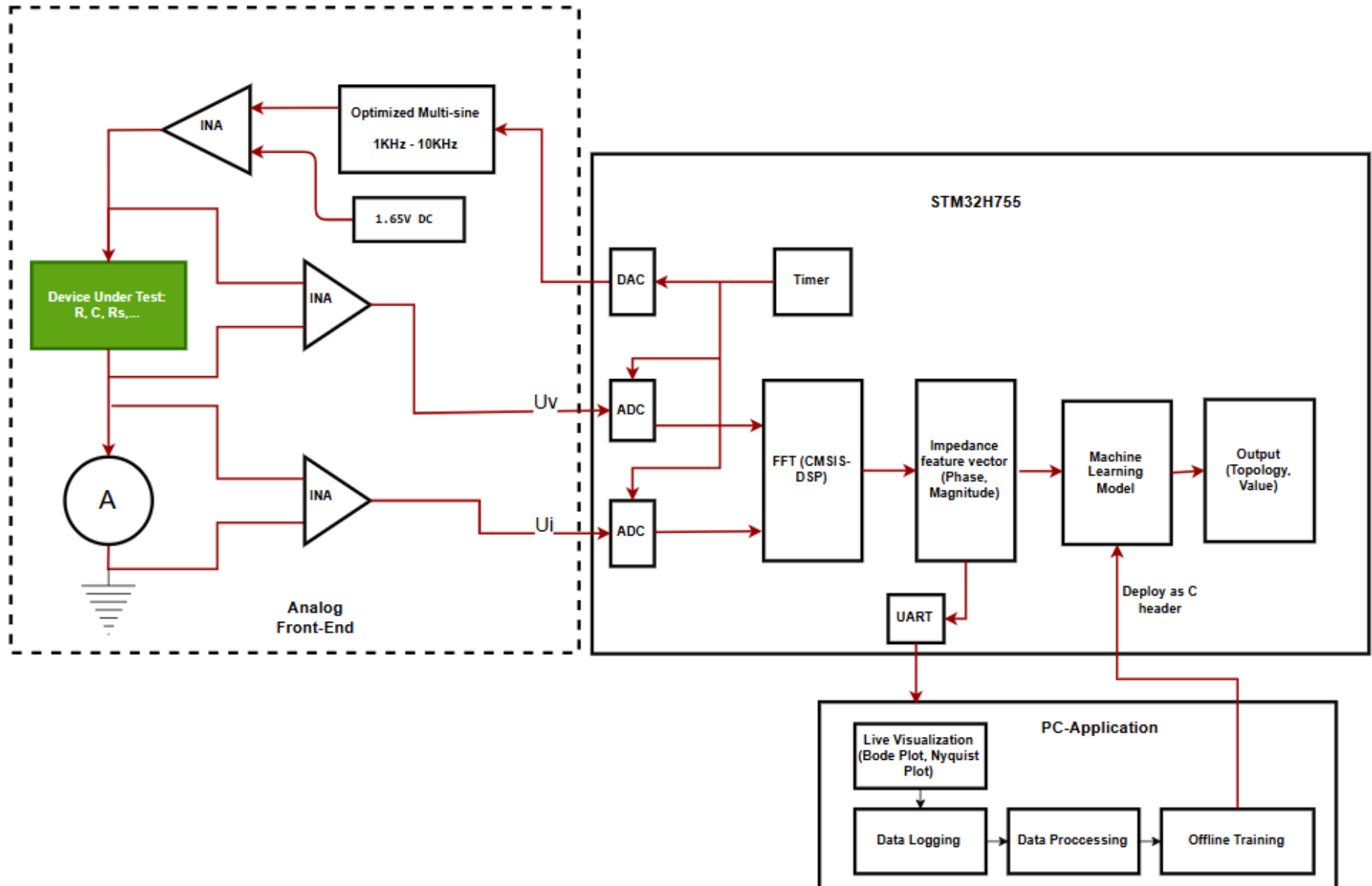
How to acquire a fast, sufficiently accurate impedance spectrum using only peripherals already built into a general-purpose MCU?



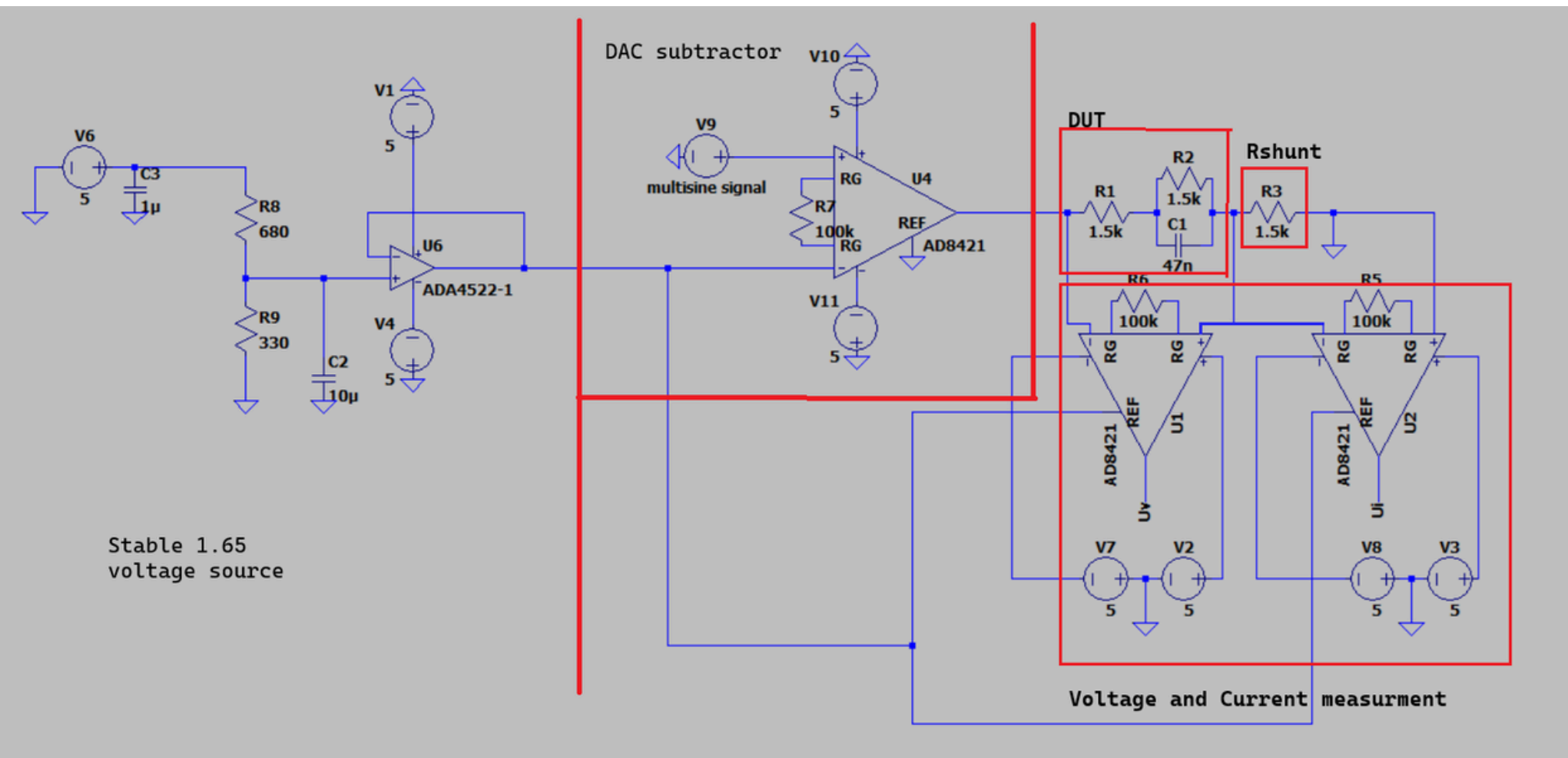
Q2 — Inference

How to recover circuit topology and component values on-device, within a few tens of kB of RAM and a few ms of runtime?

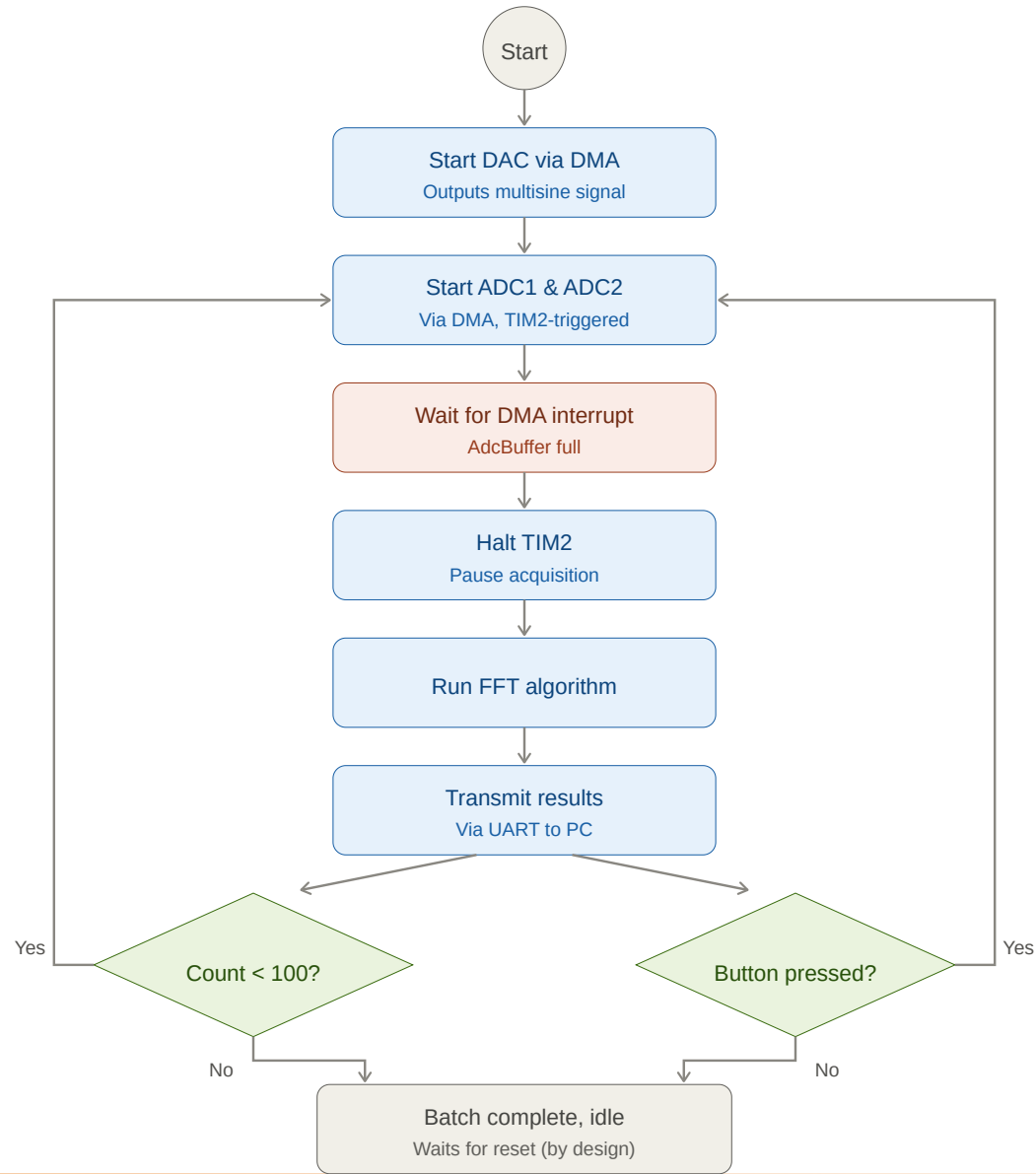
Overview of proposed Architecture



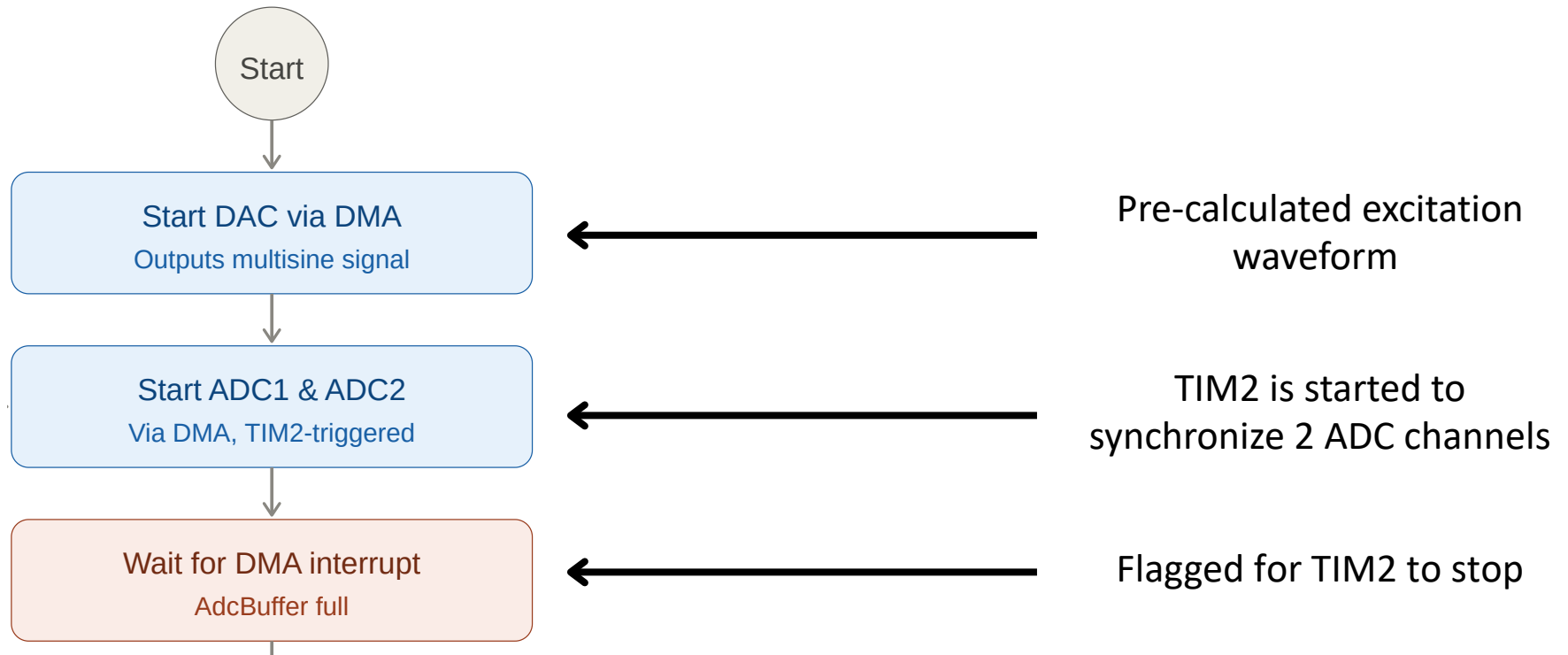
Hardware — Analog Front-End



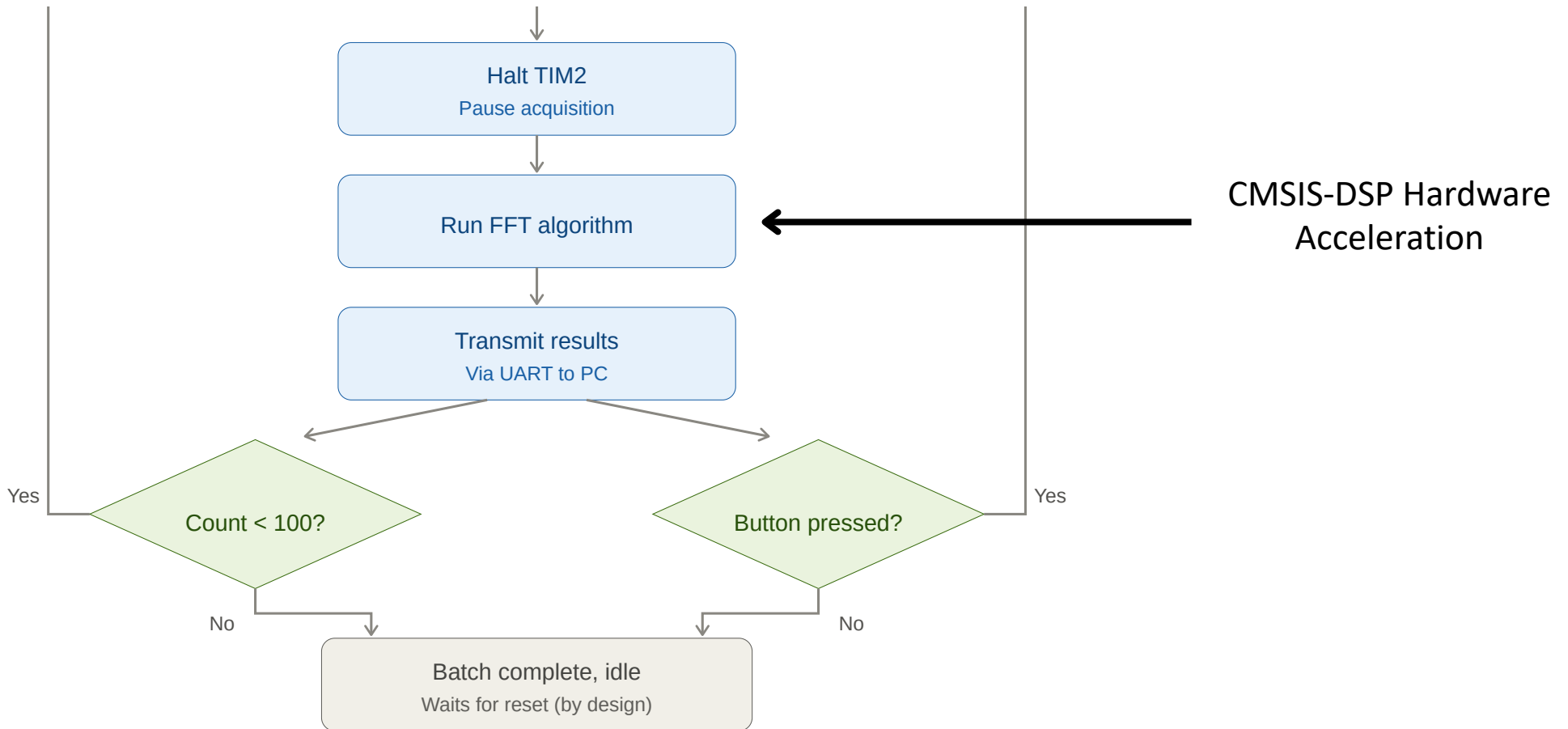
Firmware & DSP



Firmware & DSP



Firmware & DSP



IS Measurement Results

Test Setup

- 4 topologies: R, C, Series RC, Parallel RC
- Grid: 6 R-values $\times$ 5–6 C-values
- 60–85 repeated measurements per combination

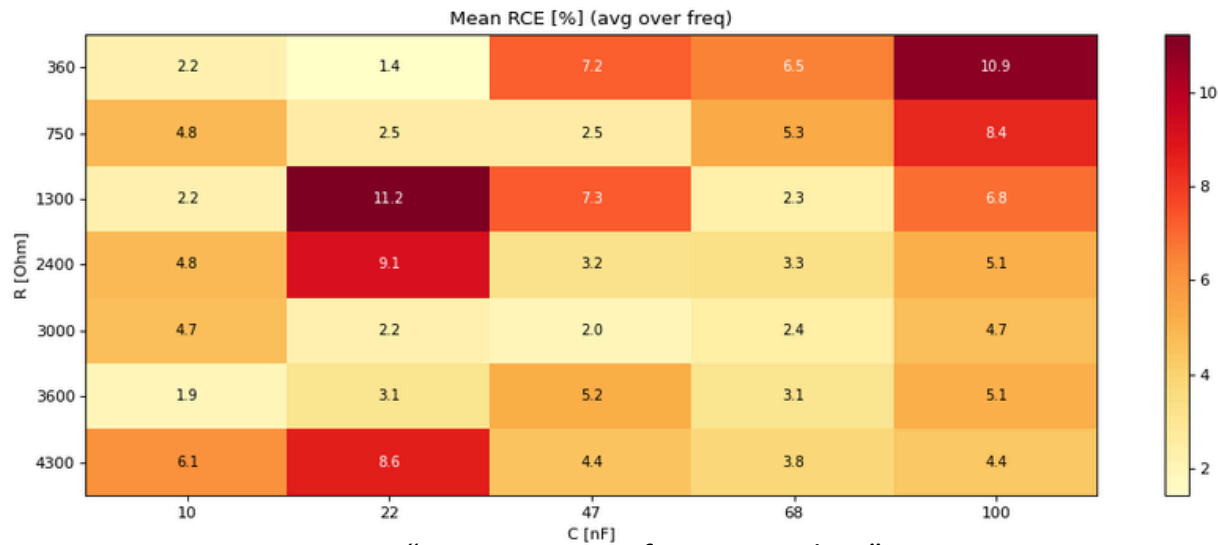
Evaluation Metric — Relative Complex Error (RCE)

- Compares measured complex impedance vs. theoretical value
- Reference computed from nominal (labeled) R, C values
- Evaluated across 1–10 kHz excitation band

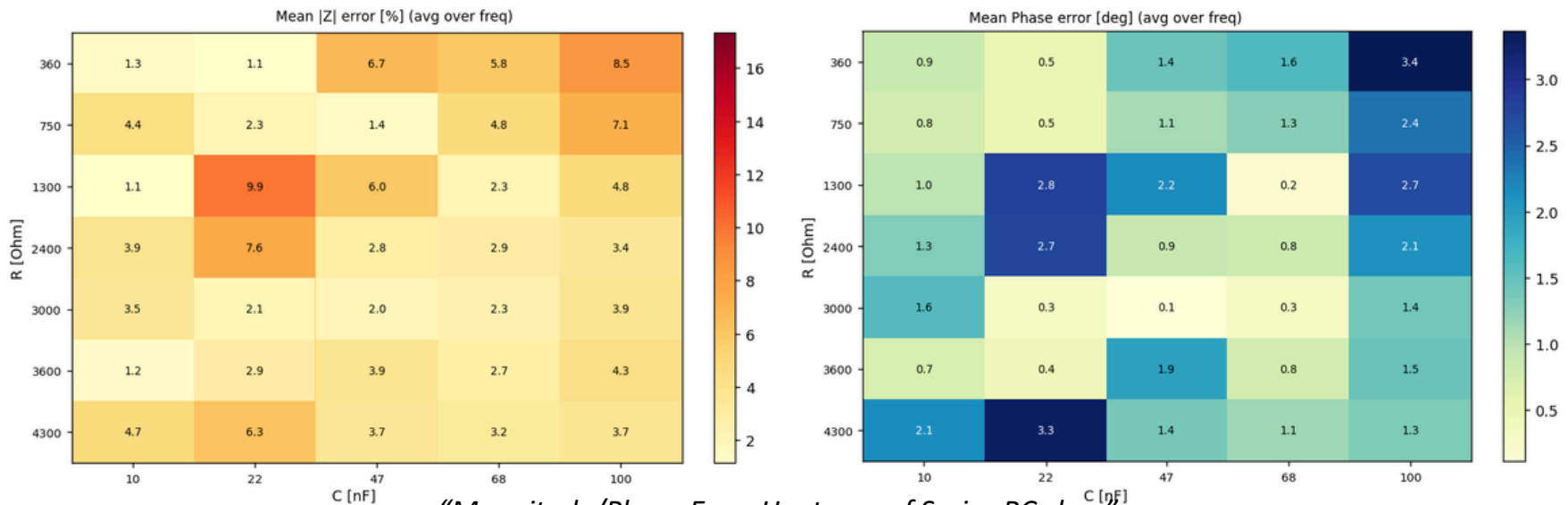
Component Class	Value (R, C)	RCE @ ~ 1 kHz [%]	RCE @ ~ 5 kHz [%]	RCE @ ~ 10 kHz [%]	Trend Observation
Resistor	1300 Ω , 0 nF	1.7 ± 0.8	1.5 ± 0.6	1.8 ± 1.0	Stable across band
Capacitor	0 Ω , 10 nF	6.5 ± 4.0	2.5 ± 1.5	3.0 ± 1.2	Improves at high freq
Series RC	2400 Ω , 22 nF	13.5 ± 3.0	6.0 ± 1.5	3.5 ± 1.0	High error at low freq
Parallel RC	750 Ω , 100 nF	4.0 ± 1.5	15.0 ± 4.0	22.0 ± 5.5	Degrades at high freq

“Relative Complex Error (RCE) at Anchor Frequencies for Representative Components”

IS Measurement Results

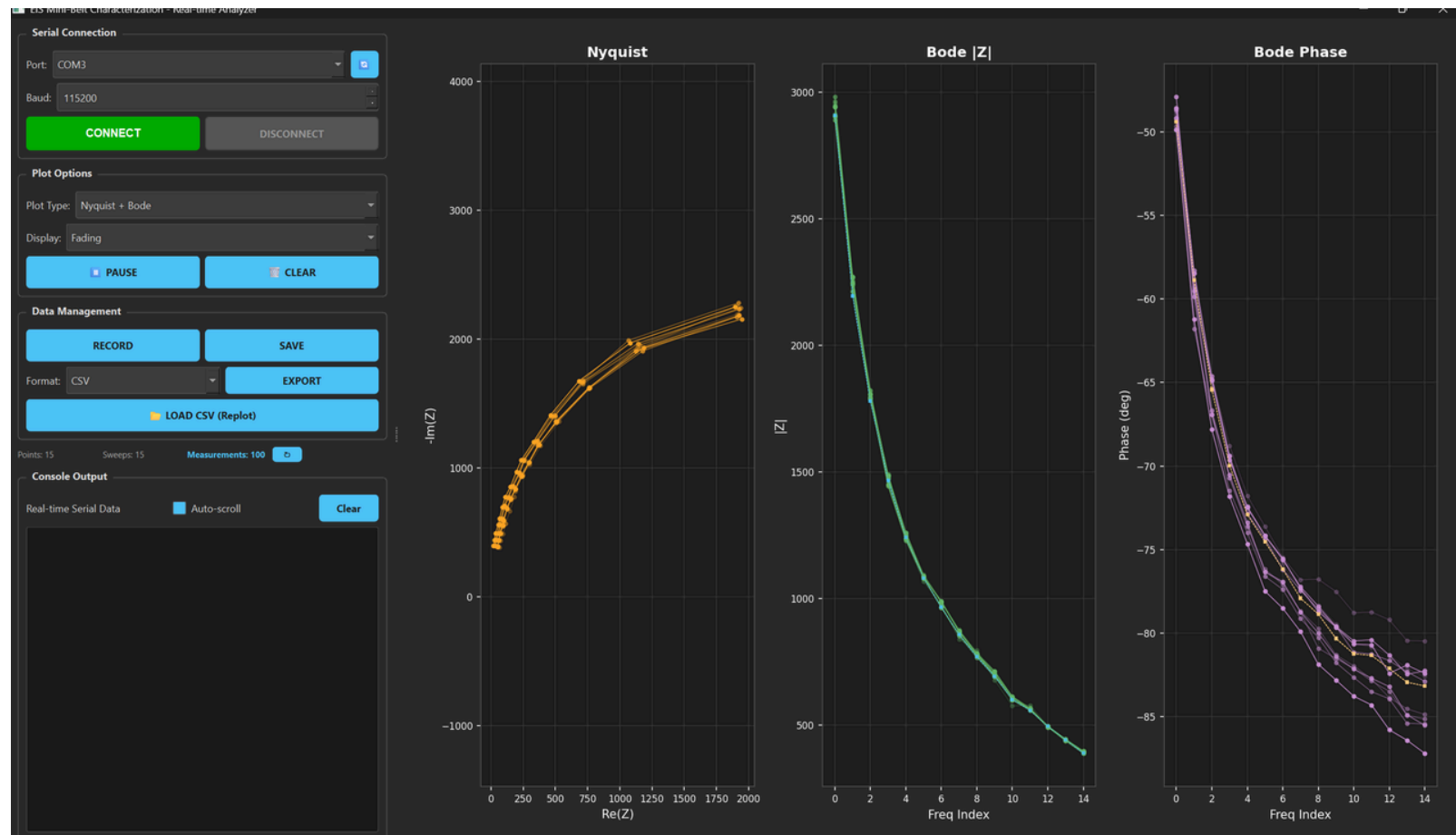


"RCE Heatmap of Series RC class"



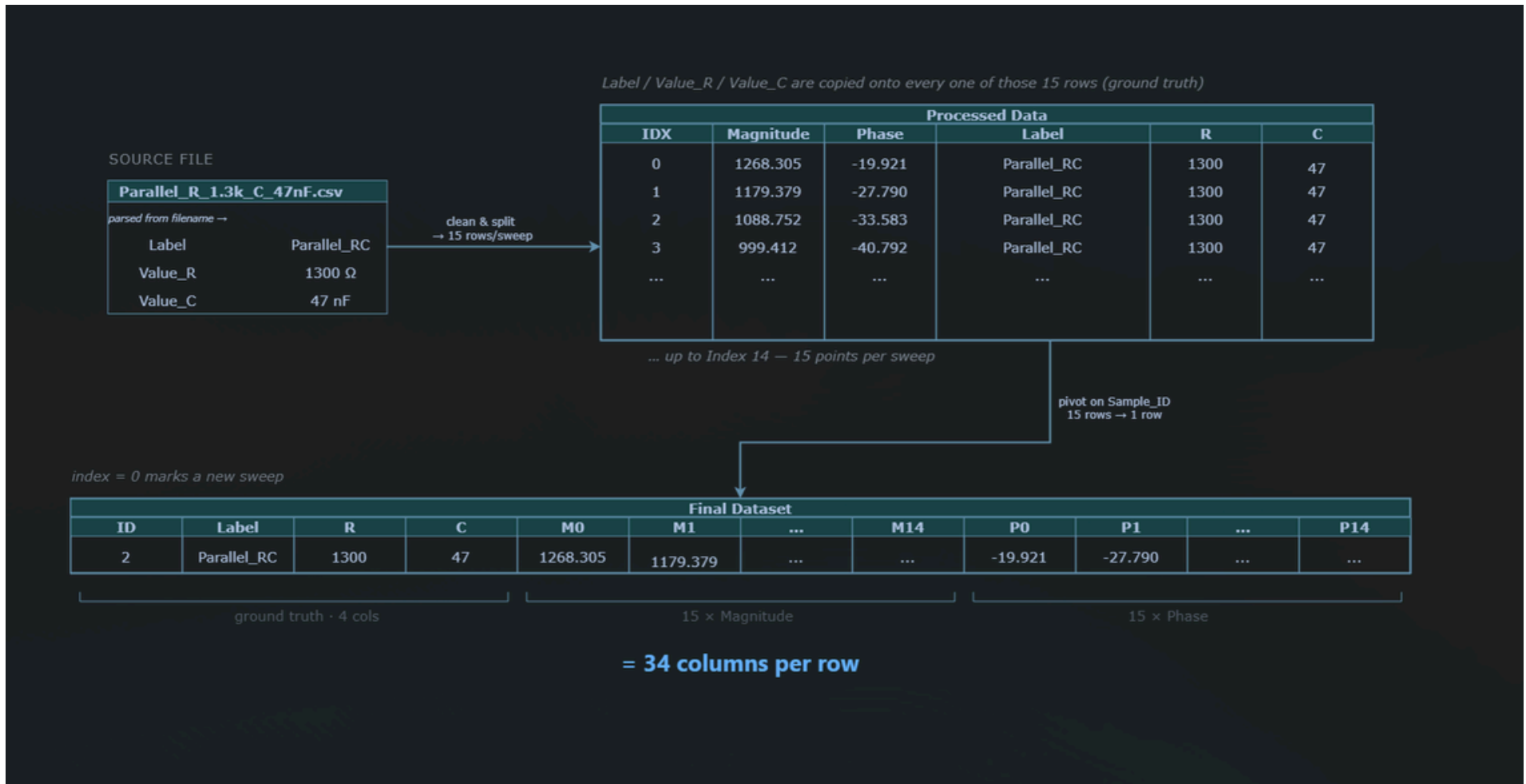
"Magnitude/Phase Error Heatmap of Series RC class"

GUI



- Receives & plots UART data in real time (Nyquist, Bode $|Z|$, Bode phase)
- Labels data (Label, R, C) + exports CSV → the source of the training data
- Its real role: a quality-control gate before entering the ML pipeline

Dataset Pipeline



Dataset Pipeline

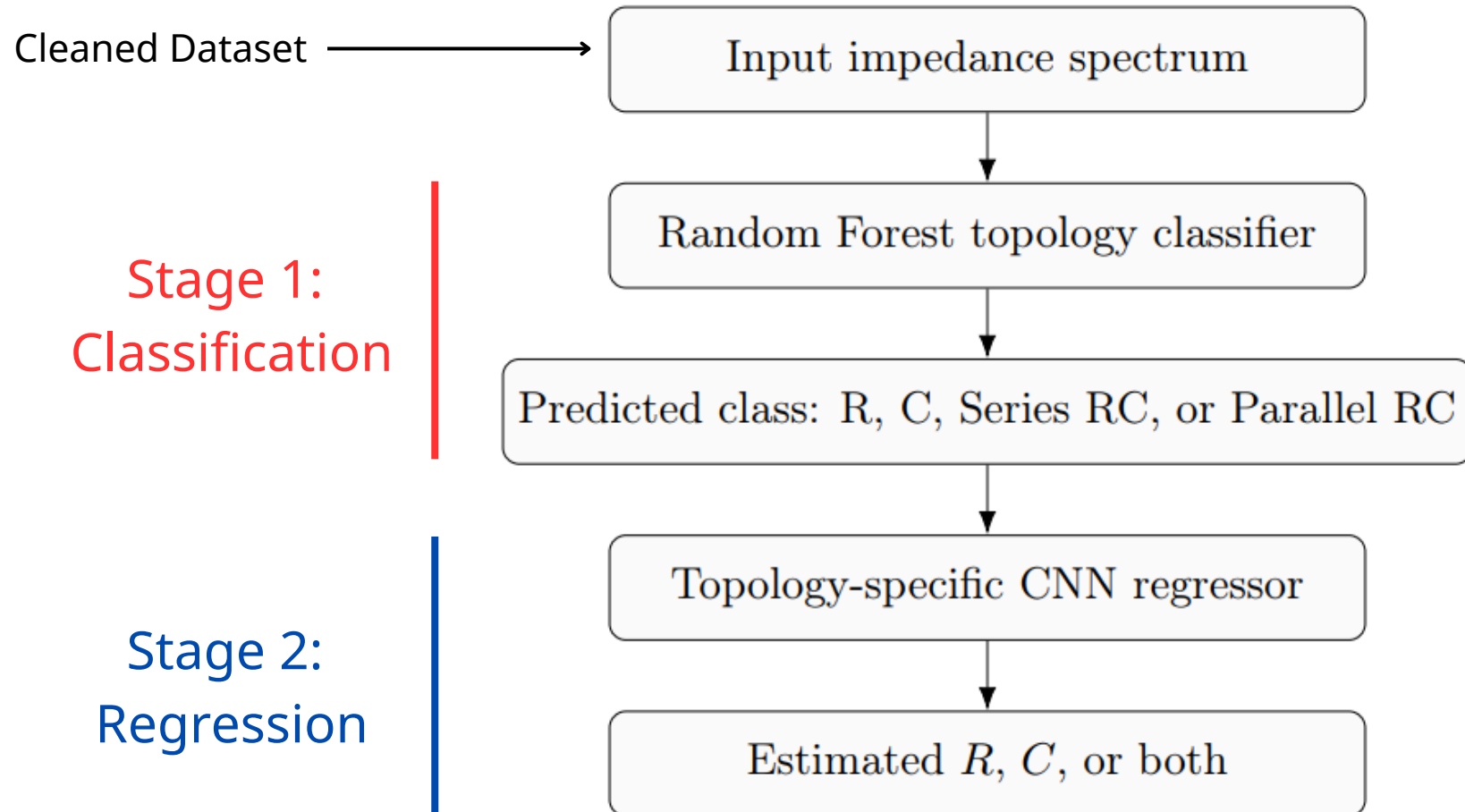
Result :

Original Filename (Mags and Phases)	Label (Topology)	Value_R (Ω)	Value_C (nF)
Parallel_R_1.3k_C_100nF.csv	Parallel_RC	1300.0	100.0
Series_R_3k_C_33nF.csv	Series_RC	3.0	33.0
Pure_C_100nF.csv	Capacitor	0.0	100.0
nothing.csv	nothing	0.0	0.0



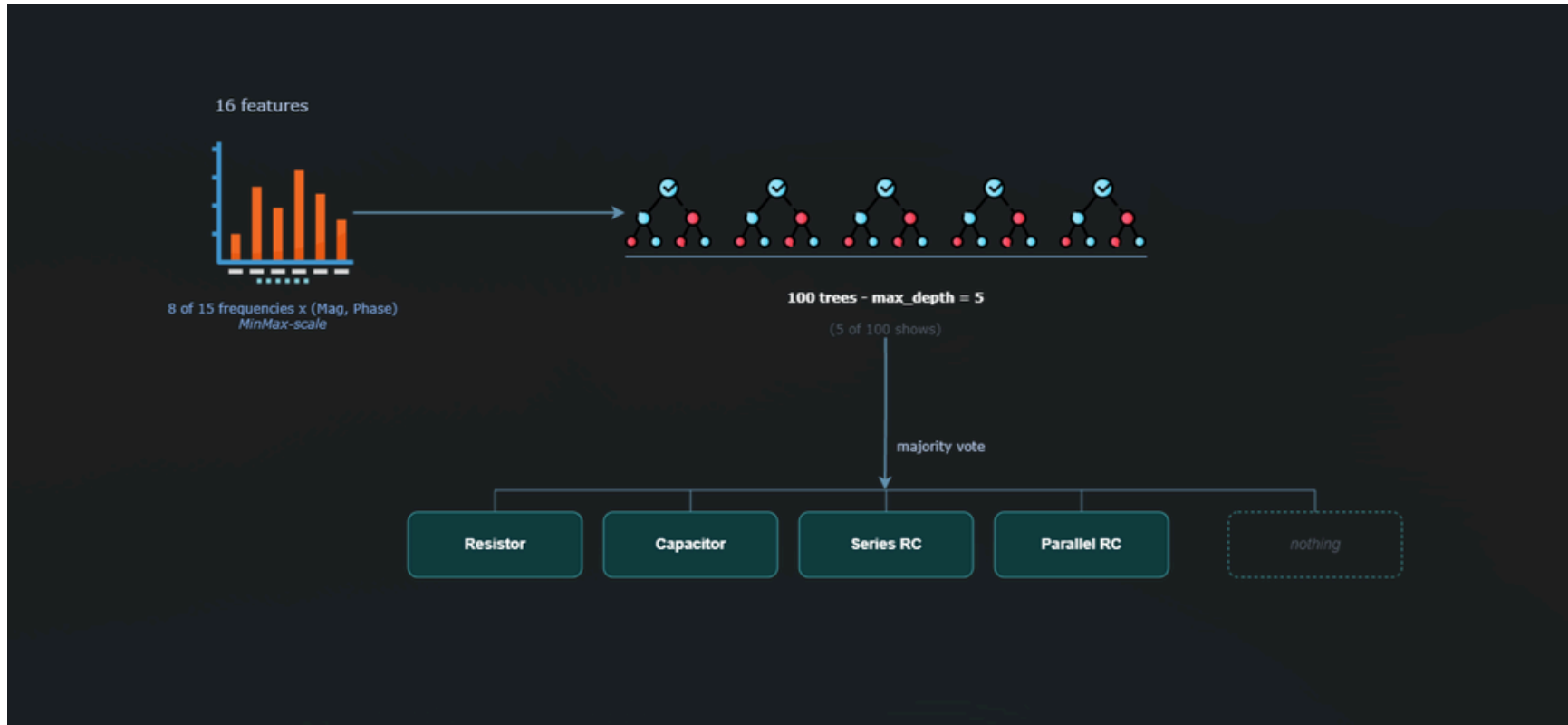
Sample_ID	Label	Value_R	Value_C	Mag_0	...	Mag_14	Phase_0	...	Phase_14
1	nothing	0.0	0.0	139 435.6	...	177 898.9	-147.19	...	-152.67
2	Parallel_RC	1300.0	100.0	1098.359	...	191.294	-36.673	...	-82.793
3	Resistor	2400.0	0.0	2466.725	...	2450.957	0.828	...	3.667
4	Capacitor	0.0	100.0	1716.186	...	172.145	-88.41	...	-90.837

Machine Learning Model Design



Machine Learning Model Design

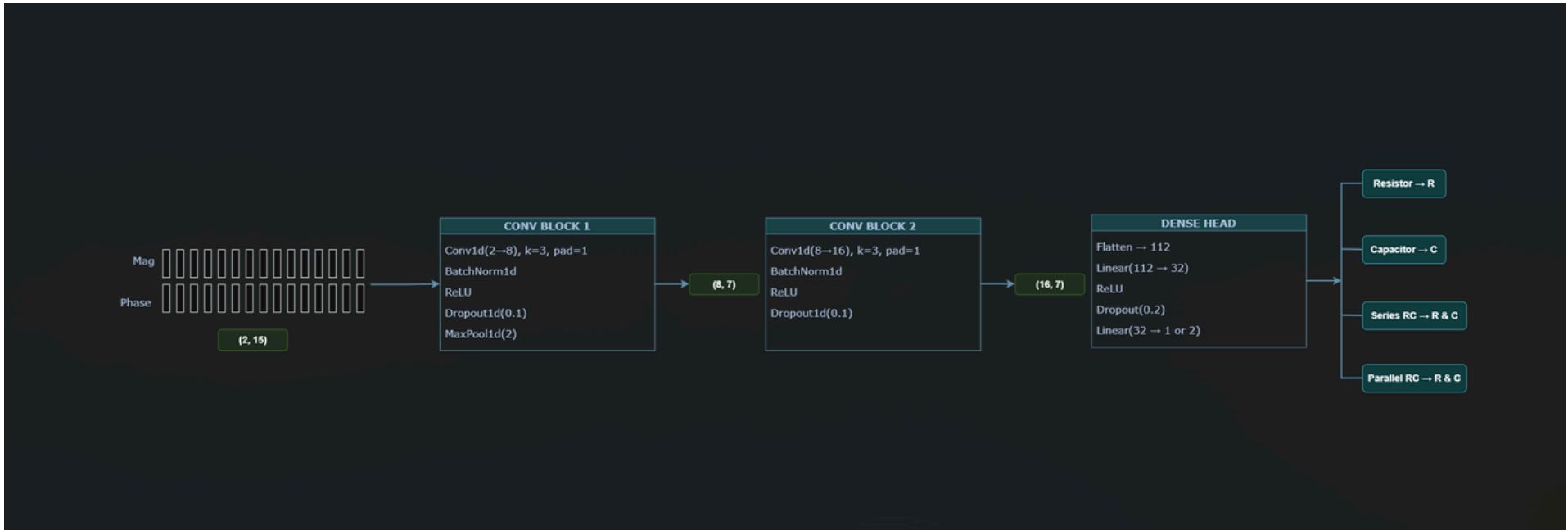
Stage 1: Classification



- Input: 16 features (magnitude + phase at 8 of 15 frequencies), MinMax-scaled
- Config: 100 trees, max_depth = 5
- Benchmark it with Decision Tree, SVM-RBF, MLP

Machine Learning Model Design

Stage 2: Regression



- Input: 2×15 tensor - a Magnitude channel and a Phase channel over frequency
- 4 independent instances - trained separately per topology
- Split: GroupShuffleSplit (80/20) - scalars fit only on the training set
- Optimizer AdamW · Huber loss (robust to outliers) · LR scheduler · Early stopping

Results and Evaluation

Classification

Model (Overall Accuracy)	Class	Precision	Recall	F1-score	Support
Decision Tree (98.46%)	Capacitor	0.87	0.96	0.91	180
	Parallel_RC	1.00	1.00	1.00	1080
	Resistor	0.98	0.99	0.98	180
	Series_RC	0.99	0.98	0.99	1260
	nothing	1.00	1.00	1.00	30
Random Forest (98.75%)	Capacitor	0.88	0.98	0.93	180
	Parallel_RC	0.99	1.00	1.00	1080
	Resistor	1.00	0.96	0.98	180
	Series_RC	1.00	0.98	0.99	1260
	nothing	1.00	1.00	1.00	30
SVM (RBF Kernel) (96.56%)	Capacitor	0.79	0.99	0.88	180
	Parallel_RC	0.96	1.00	0.98	1080
	Resistor	0.97	0.81	0.88	180
	Series_RC	1.00	0.96	0.98	1260
	nothing	1.00	1.00	1.00	30
MLP (Perceptron) (98.28%)	Capacitor	0.87	0.97	0.92	180
	Parallel_RC	0.99	1.00	0.99	1080
	Resistor	1.00	0.91	0.95	180
	Series_RC	1.00	0.98	0.99	1260
	nothing	1.00	1.00	1.00	30

Results and Evaluation

Regression

Routing Mode	Topology	MAE_R (Ω)	MAE_C (nF)	Support
Oracle (upper bound)	Capacitor	0.00	10.27	190
	Parallel_RC	196.80	3.66	855
	Resistor	162.03	0.00	95
	Series_RC	171.19	5.52	570
End-to-end (deployed)	Capacitor	0.00	10.27	190
	Parallel_RC	196.80	3.66	855
	Resistor	162.03	0.00	95
	Series_RC	174.74	5.62	570

Table 14: Overall MAE

Routing Mode	Overall MAE_R (Ω)	Overall MAE_C (nF)	Misclassified
Oracle routing	185.03	5.09	—
End-to-end routing	186.36	5.13	6 / 1710 (0.35%)
Routing gap	+1.33	+0.04	

Results and Evaluation

Deployment

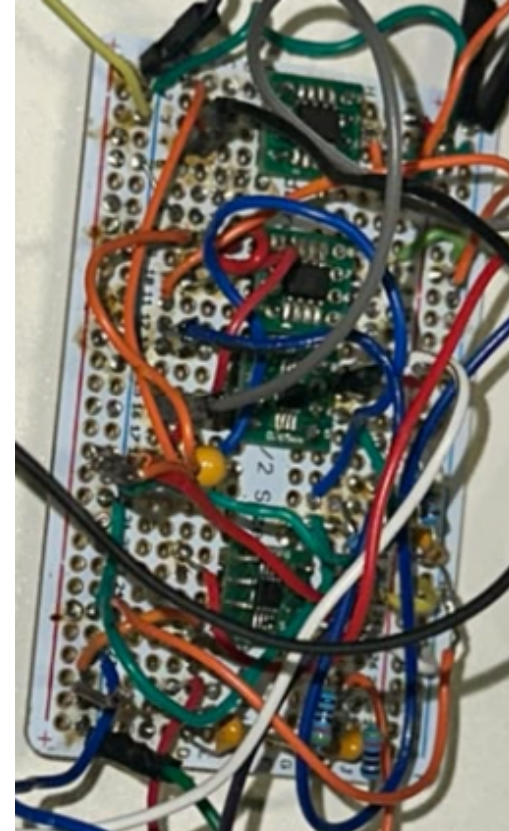
Flash (Before - After)	80,29 kB (3,92%) - 358,02 kB (17,48%)
RAM (Before - After)	35,54 kB (3,47%) - 35,63 kB (3,48%)
WCET	13057,48 us

=> The machine learning model needs:

- 13,56 % of total Flash
- 0,01 % of total RAM
- about 13 ms of runtime per iteration

Limitations

- **Self-soldered wiring**
=> messy circuit,
unwanted noise,
hard for rewiring
- **Use manufacturer's
values as baseline**
=> not very accurate
for circuit and ML
model analysis



Limitations

- **Imbalanced dataset**
=> varying reliability between classes
- **Data splitting strategy**
=> random split in classification stage allow similar impedance patterns in both training and validation → Memorizing but not generalizing

Label	# sample
Series_RC	3,005
Parallel_RC	2,674
Capacitor	437
Resistor	391
nothing	93
Total	6,600

Limitations

- **Error propagation**

=> wrong classification surely leads to wrong regression

- **Limited number of circuit configurations**

=> still cannot work with other more complex circuits

- **Logarithmic error scaling**

=> small deviation in regression results can lead to large deviation in R/C values

$$\boxed{y}^{\text{R/C value}} = 10^{\boxed{y'}^{\text{regression result}}}$$

Conclusions

- System is usable, accurate, fast and resource-efficient.
- Some possibilities for improvements:
 - Design a dedicated board for the system (PCB)
 - Apply more advanced calibration techniques
 - Train the ML model with more complex configurations
 - Collect more data for training

Thank you for attention

Q&A